

The PTA Likelihood

Background

Everything in data analysis begins with a simple equation:

$$\text{data} = \text{signal} + \text{noise}.$$

What you call “signal” is up to you. It could be caused by GWs, Dark Matter, Pulsar Physics, etc. **It does not have to be deterministic either. A Signal could itself be a type of noise.**

What you call “noise” is often not up to you! Your data gathering instruments dictate what noise is. It often is related to **measurement uncertainty. It is ok if noise itself is not fully understood.** As long as you have a model for it, you are good!

In PTAs, we promise ourselves that our modeling of “signal” is so advanced that “noise” is simply a series of random numbers drawn from a Gaussian distribution with zero mean and **an unknown variance, N :**

$$\text{noise at time } t_i \sim \text{Normal}(\text{mean} = 0, \text{var} = N).$$

Everything non-Gaussian is promised to be captured by “signal” or ignored because, in practice, it makes no difference to include them or not.

So, if

$$\text{data} = \text{signal} + \text{noise},$$

then

$$\text{noise} = \text{data} - \text{signal}.$$

Also, if

$$\text{noise}|_{t_i} \sim \text{Normal}(\text{mean} = 0, \text{var} = N),$$

then,

$$(\text{data} - \text{signal})|_{t=t_i} \sim \text{Normal}(\text{mean} = 0, \text{var} = N).$$

In other words, we can write generally that

$$p(\text{data} - \text{signal}) = \frac{\exp\left\{-\frac{1}{2}\left[(\text{data} - \text{signal})^T N^{-1}(\text{data} - \text{signal})\right]\right\}}{\sqrt{\det(2\pi N)}}.$$

This is the general form of the PTA likelihood. Pretty simple! The devil is in the details, though!

What is data and signal?

For PTAs, the data is the sequence of observed TOAs:

$$\begin{aligned}\text{data} &= t_{\text{obs}}, \\ \text{signal} &= t_{\text{timing model}} + \delta t.\end{aligned}$$

The timing model describes almost all of the data. It is the dominant source of the PTA data (if you paid attention on Tuesday, you should not be surprised!). To make matters worse for non-timing model sources, the timing model itself is not well understood. **Its model parameters are uncertain. So, even the small δt is not purely made out of sources outside the timing model.**

is where the fun is! So, let's model it:

$$\delta t = \underbrace{\text{GW Effects}}_{\substack{\text{*Gravitational Wave Background} \\ \text{*Individual Continuous Waves} \\ \text{*Burst}}} + \text{Pulsar Spin Noise Effects} + \underbrace{\text{Other}}_{\substack{\text{*Interstellar random} \\ \text{effects such as} \\ \text{unmodeled DM,} \\ \text{solar wind, etc} \\ \text{*Other things you like!}}}$$

How do we model δt mathematically?

Everything that is considered indeterministic (i.e., random) is written as a product of a basis and a set of coefficients:

$$\delta t_{\text{random}} = \underset{\text{basis matrix coefficients}}{B} b$$

The B basis matrix has the shape that is $(n_{\text{TOAs}}, \text{size of } b)$. The vector b is just a representation of δt_{random} in another domain compared to time. B projects b to the time domain from another domain.

Why change domain?

- 1) Computational efficiency. The time domain is huge. It can be reduced to a much smaller domain
- 2) Often, the models behind astrophysical/cosmological signals explain things in the frequency domain. The frequency domain is just more interesting/informative with respect to most models.

Example:

- 1) For GWs, we use the Fourier basis. So B becomes F , and b becomes a . This is just another way of saying that **we model things in the frequency domain, then move those things from the frequency to the time domain using the discrete Fourier transform. Thus, mathematically,**

$$\delta t_{\text{GW}} = Fa.$$

This basis can be used for all GW signals, including those that are deterministic!

- 2) To account for the uncertainty in the timing model parameters, a special basis is constructed that encodes the general form of the signal induced by the timing model parameters. This basis is called M , the design matrix.

$$\delta t_{\text{timing model uncert}} = M \varepsilon.$$

ε is the uncertainty parameter. It is supposed to be very tiny if the timing model parameters are close to their true values. You can think of M as being a polynomial basis matrix.

NOTE: you have all the power in the world to choose your bases. These M and F are just more common than other bases.

At the end, we have

$$\begin{aligned} \text{data} &= t_{\text{obs}} \\ \text{signal} &= t_{\text{timing model}} + M \varepsilon + (Fa)_{\text{GWB}} + (Fa)_{\text{CW}} + (Fa)_{\text{DM}} + \dots \end{aligned}$$

What is the PTA likelihood?

$$p(t_{\text{obs}} | \text{our model}) = \frac{\exp\left\{-\frac{1}{2}\left[(t_{\text{obs}} - r)^T N^{-1} (t_{\text{obs}} - r)\right]\right\}}{\sqrt{\det(2\pi N)}}.$$

$$r = t_{\text{timing model}} + M\varepsilon + (Fa)_{\text{GWB}} + (Fa)_{\text{CW}} + (Fa)_{\text{DM}} + \dots$$

$$N \approx \begin{bmatrix} \sigma_{\text{first-obs}}^2 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & \sigma_{\text{last-obs}}^2 \end{bmatrix}$$

$$\sigma_i^2 \equiv \underbrace{\text{EFAC}^2}_{\text{telescope-specific parameter}} \left(\sigma_{\text{measured-uncert at } i}^2 + \underbrace{\text{EQUAD}}_{\text{telescope-specific parameter}} \right)$$

PTA Prior

Background

Remember,

$$\underbrace{p(\text{model} | \text{data})}_{\text{The goal (posterior)}} \propto \underbrace{p(\text{data} | \text{model})}_{\text{likelihood}} \underbrace{p(\text{model})}_{\text{prior}}.$$

The PTA likelihood on its own does not give you everything you need. **With it, you describe how the output of a given model can explain the data.** However, it says nothing about the model itself:

- Are all the model parameters independent?
- Are they bounded (we cannot have negative mass!)?
- Do you expect some values of model parameters to be more probable than others?

Remember, inference is about updating prior information using data. You **MUST** have some prior information to begin with. It could be very minimal, but it has to be something. By changing your model, you change your prior belief. The whole GW literature in PTAs is about messing with priors.

Prior Choices

Category 1: (Common) Assumptions

This is the most debatable part of PTAs. People just do not like assumptions that others make about certain models. The most common ones are:

- In the $\delta t_{\text{random}} = Fa$ modeling, a from different frequencies are assumed to be independent.
 - Is this assumption theoretically and strictly right? No!
 - Does it matter in practice? Depending on the model you are considering, the answer can be either yes or no. For astrophysically motivated models, the answer is not yet explored.
 - ***RANT: **no one should care about coefficients** . What matters is understanding the source behind the signal that tries to model. Just because an assumption affects the shape of the distribution of does not mean it affects how much we will learn about the source with or without the assumption. The data analysis problem of PTAs is a lot more complicated than that! End of RANT***.
- The timing model uncertainty is truly tiny. So, in $\delta t_{\text{timing model uncert}} = M\varepsilon$, ε is assumed to be tiny. If your timing model is highly non-linear or the values of timing model parameters are way off of their true values, this assumption breaks!
- The variance of ε parameter is infinitely large. In other words, we know jack ... about it, a-priori.

Category 2: Hierarchical relationships

Do you see any mass of graviton in the PTA likelihood? Do you see anything about SMBHBs? So, how do we infer such things if they are not in the likelihood explicitly? The answer is via hierarchical priors. For example,

$$\underbrace{p\left(a_{\text{GWB}} \mid \underbrace{h_c^2}_{\text{char-strain}}\right)}_{\text{hierarchical prior}} = N\left(0, \underbrace{g(h_c^2)}_{\text{a function}}\right).$$

So, to make a_{GWB} under your model, you need to know h_c^2 (a hierarchical relationship!) . Hierarchical priors are everywhere in PTAs.

Category 3: Marginal Priors

You need at least one marginal (non-hierarchical) prior in your model. Something that everything else builds from it. In the example

$$\underbrace{p\left(a_{\text{GWB}} \mid \underbrace{h_c^2}_{\text{char-strain}}\right)}_{\text{hierarchical prior}} = N\left(0, \underbrace{g}_{\text{a function}}(h_c^2)\right),$$

you still need to have a prior probability for h_c^2 . To complete the prior information, you need to attach a marginal prior to the rest of the hierarchical priors.

So the complete version of above is

$$p(a, h_c^2) = \underbrace{p(h_c^2)}_{\text{marginal prior}} p(a_{\text{GWB}} \mid h_c^2).$$

In general, you can write

$$\begin{aligned} \underbrace{p\left(a, \overbrace{\alpha, \beta, \gamma, \dots}^{\text{model params}}\right)}_{\text{total prior}} &= \underbrace{p(a)}_{\text{marginal prior}} [p(a \mid \alpha, \beta, \gamma, \dots)] [p(\alpha, \beta, \gamma, \dots)] \\ &= p(a) [p(a \mid \alpha, \beta, \gamma, \dots)] [p(\alpha \mid \beta, \gamma, \dots)] [p(\beta, \gamma, \dots)] \\ &= p(a) [p(a \mid \alpha, \beta, \gamma, \dots)] [p(\alpha \mid \beta, \gamma, \dots)] [p(\beta \mid \gamma, \dots)] [p(\gamma, \dots)] \\ &= p(a) [p(a \mid \alpha, \beta, \gamma, \dots)] [p(\alpha \mid \beta, \gamma, \dots)] [p(\beta \mid \gamma, \dots)] [p(\gamma \mid \dots)] [p(\dots)] \end{aligned}$$

Notice how one layer is added in each step. You can keep doing this in your model until you run out of model parameters!

Completing the PTA Puzzle

Knowing both the likelihood and the prior for a given model means you are done! All you need to do is to let the computers do the inference for you via MCMC. All MCMC algorithms require only these two. Give them the PTA likelihood, your

prior for your model, and they will do the rest. Let's do it! See the Jupyter notebook.

The Marginalized Likelihood

Well, that did not work. Why? We are varying the coefficients alongside their variance. It is too much for MCMCs to handle. So, what is the cure? Marginalizing over the Fourier coefficients!

$$p(h_c^2 | t_{\text{obs}}) = \int da_{\text{GWB}} \left\{ \underbrace{\left[p(h_c^2) p(a_{\text{GWB}} | h_c^2) \right]}_{\text{prior}} \underbrace{\left[\frac{\exp \left\{ -\frac{1}{2} \left[(\delta t_M - F a_{\text{GWB}})^T N^{-1} (\delta t_M - F a_{\text{GWB}}) \right] \right\}}{\sqrt{\det(2\pi N)}} \right]}_{\text{likelihood}} \right\}$$

What is the answer?

$$\begin{aligned} p(h_c^2 | t_{\text{obs}}) &= N(0, C) \\ &= \frac{\exp \left\{ -\frac{1}{2} \left[(\delta t_M)^T C^{-1} (\delta t_M) \right] \right\}}{\sqrt{\det(2\pi C)}} \\ C &= F \varphi F^T + N \\ \varphi &= \begin{bmatrix} h_{c;0}^2 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & h_{c;\text{last-freq}}^2 \end{bmatrix} \end{aligned}$$

Note that the marginalized likelihood incorporates a part of the prior in it. So, it is not really a likelihood in the purest sense of the word. But it is a common terminology in PTA literature.

Questions

1. The likelihood just above is for a single pulsar. What is the likelihood for a multi-pulsar PTA? HINT: look at the jupyter notebook carefully.
2. The Hellings and Downs correlations are missing. Where would you add them? What quantity and where?

3. In the jupyter notebook, increase the number of pulsars by 1, up to 4 pulsars. Save the posteriors. Compare the posteriors.
4. Can you recover what you inject? Is there a problem?